MATHEMATICS

GENERAL NOTATION, SETS AND PROBABILITY

General notation

Symbol	Meaning	Examples
$=$ $\neq$	Equal Not equal	$2+3=5$; $4-4=0$ $2\neq 3$; $-1\neq 1$
$\approx$	Approximately	$\pi\approx 3.14$ $\sqrt{2}\approx 1.41$
$<$ $>$	Less than Greater than	$2<3$; $-1<0$ $5>4$; $0>-2$
$\leq$ $\geq$	Less than or equal Greater than or equal	$2\leq 2$; $-1\leq 3$ $5\geq 4$; $0\geq 0$
$\pm$	Plus or minus	$3\pm 1=4$ or 2 $-2\pm 3=1$ or -5
$\lvert a\rvert$	Absolute value	$\lvert -5\rvert=5$; $\lvert 3\rvert=3$
$\sqrt{a}$	Square root	$\sqrt{9}=3$; $\sqrt{\frac{1}{4}}=\frac{1}{2}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set with three elements	$\{2,4,6\}$; $\{-1,0,5\}$
$\varnothing$	Empty set	$\{1\}\cap\{2\}=\varnothing$ $\{x\in\mathbb{R}: x^2=-1\}=\varnothing$
$\in$ $\notin$	Is element of Is not element of	$2\in\{1,2\}$; $-1\in\mathbb{Z}$ $3\notin\{1,2\}$; $\frac{1}{2}\notin\mathbb{Z}$
$\subseteq$	Is a subset of	$\{1\}\subseteq\{1,2\}$ $\mathbb{N}\subseteq\mathbb{Z}$
$\cup$ $\cap$	Union Intersection	$\{1\}\cup\{2\}=\{1,2\}$ $\varnothing\cup\{0\}=\{0\}$ $\{1,2\}\cap\{2\}=\{2\}$ $\{-1\}\cap\{4\}=\varnothing$
$\mathbb{N}$ $\mathbb{Z}$	Natural numbers Integers	$3\in\mathbb{N}$; $8\in\mathbb{N}$ $-2\in\mathbb{Z}$; $0\in\mathbb{Z}$
$\mathbb{Q}$ $\mathbb{R}$	Rational numbers Real numbers	$\frac{1}{2}\in\mathbb{Q}$; $-3\in\mathbb{Q}$ $\sqrt{2}\in\mathbb{R}$; $\frac{1}{4}\in\mathbb{R}$
(a,b) $[a,b]$	Open interval Closed interval	$1\notin(1,3)$; $0\in(-2,2)$ $1\in[1,3]$; $-2\in[-2,0]$

Probability and events

Example contexts
I: fair die **II**: two independent fair coin tosses.
H = heads; T = tails.
I: $A=\{1,3,5\}$, $B=\{1,2\}$.
II: $A=\{\text{HH}\}$, $B=\{\text{HH},\text{HT},\text{TH}\}$.

Symbol	Meaning	Examples
Ω	Sample space: all possible outcomes	I: $\Omega=\{1,2,3,4,5,6\}$ II: $\Omega=\{\text{HH,HT,TH,TT}\}$
ω	An individual outcome	I: $\omega=3\quad\in\Omega$ II: $\omega=\text{HT}\in\Omega$
$A=\{1,3,5\}$	Event shown as a set of outcomes	I: $A=\{1,3,5\}$: odd II: $A=\{\text{HH}\}$: two heads
$A\subseteq\Omega$	An event is a subset of the sample space	I: $\{1,3,5\}\subseteq\Omega$ II: $A\subseteq\Omega$
$P(A)$	Probability of event A	I: $P(A)=\frac{3}{6}=\frac{1}{2}$ II: $P(A)=\frac{1}{4}$
A^c	Complement of event A	I: $A^c=\{2,4,6\}$ II: $A^c=\{\text{HT},\text{TH},\text{TT}\}$
$A\cup B$	Event A or B (or both)	I: $A\cup B=\{1,2,3,5\}$ II: $A\cup B=\{\text{HH,HT,TH}\}$
$A\cap B$	Both events A and B	I: $A\cap B=\{1\}$ II: $A\cap B=\{\text{HH}\}$

Symbols acquire meaning from their mathematical context.

MATHEMATICS

FUNCTIONS, VARIABLES, SEQUENCES AND SUMS

Function notation

Example contexts
I: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ **II**: $g : [0, \infty) \to \mathbb{R}$, $g(x) = \sqrt{x}$

Symbol	Meaning	Examples
f	Name of a function	I: $f(x) = x^2$ II: $g(x) = \sqrt{x}$
$f(x)$	Value of f at input x	I: $f(3) = 3^2 = 9$ II: $g(4) = \sqrt{4} = 2$
$f : A \to B$	Function from set A to set B	I: $f : \mathbb{R} \to \mathbb{R}$ II: $g : [0, \infty) \to \mathbb{R}$
$x \overset{f}{\mapsto} f(x)$	Input-to-output mapping	I: $x \overset{f}{\mapsto} x^2$; $3 \overset{f}{\mapsto} 9$ II: $x \overset{g}{\mapsto} \sqrt{x}$; $4 \overset{g}{\mapsto} 2$
$\mathrm{dom}(f)$	Domain of a function	I: $\mathrm{dom}(f) = \mathbb{R}$ II: $\mathrm{dom}(g) = [0, \infty)$

Letters and Greek symbols

Symbol	Name / description
$a, b, c, \ldots$	Common lowercase Latin letters
$A, B, C, \ldots$	Common uppercase Latin letters
x, y, z	Other common Latin letters
$\alpha,\ \beta,\ \gamma$	Alpha; beta; gamma
$\delta,\ \Delta$	Delta; uppercase delta
$\theta,\ \Theta$	Theta; uppercase theta
$\lambda,\ \Lambda$	Lambda; uppercase lambda
$\mu,\ \sigma$	Mu; sigma
$\omega,\ \Omega$	Omega; uppercase omega

Variable Examples

A variable is a symbol that can represent different values. Its allowed values and meaning depend on the context.

Possible values in separate cases

Each choice below is a separate case.

Ordinary numbers: x can be -2, 0, or 3.5.

Integers: n can be $1, 2, 3, \ldots$ when counting objects.

Angles: θ can be $30°$, $60°$, or $90°$.

Sets: A can be $\{1, 2\}$ or $\{0, 3, 5\}$.

Greek letters as numbers: λ can be -1, 0, or 2.

Letters do not have permanently assigned types. Uppercase letters often represent sets, and Greek letters often represent angles or numbers, but the context determines their meaning.

Lists, sequences and subscripts

A **sequence** is an ordered list of terms. A **series** is a sum of terms. Write sequences with parentheses and sets with braces.

Terms have positions

The terms $a_1, a_2, a_3, \ldots, a_n$ occupy positions $1, 2, 3, \ldots, n$. The notation a_i means **the term at position** i. The subscript i is an **index**, not multiplication or an exponent.

a_1 = first term
a_2 = second term
a_n = term at position n

One finite sequence, three forms

An explicit list, its indexed terms and an indexed family can describe the same sequence:

$$(2, 4, 6, 8) = (a_1, a_2, a_3, a_4) = (a_i)_{i=1}^{4}, \quad a_i = 2i.$$

In general, $(a_i)_{i=1}^{n} = (a_1, a_2, \ldots, a_n)$.

A rule generates the terms

For $a_i = 2i$, with $i = 1, 2, 3, 4$:
$a_1 = 2(1) = 2$; $a_2 = 2(2) = 4$;
$a_3 = 2(3) = 6$; $a_4 = 2(4) = 8$.
The resulting sequence is $(2, 4, 6, 8)$.

Order and repetitions matter

$(2, 4) \neq (4, 2)$ and $(2, 2, 4) \neq (2, 4)$.
For sets, order and repetition do not matter:
$\{2, 4\} = \{4, 2\} = \{2, 2, 4\}$.

From a sequence to its sum

The sum of the terms is $a_1 + a_2 + \cdots + a_n$. Summation notation writes this more compactly.

Summation notation

Read the parts

$$\sum_{i=1}^{n} a_i$$

$\sum$: summation symbol
i: index; 1: lower bound
n: upper bound
a_i: expression being summed

The index runs through consecutive integers from the lower bound to the upper bound, **inclusive**.

$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$.

Sum the sequence above: $a_i = 2i$

$$\sum_{i=1}^{4} a_i = \sum_{i=1}^{4} 2i = 2 + 4 + 6 + 8 = 20.$$

Other summands and bounds

Constant: $\sum_{i=2}^{4} 3 = 3 + 3 + 3 = 9.$

Index expression: $\sum_{i=1}^{2} (i + 3) = 4 + 5 = 9.$

Squares: $\sum_{k=1}^{3} k^2 = 1 + 4 + 9 = 14.$

The index name can change

$\sum_{i=1}^{4} i = \sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10.$
Use the new index name consistently in the bounds and summand.